

Arnav Bhatia

arnavbhatia-portfolio.vercel.app

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EDUCATION

Arizona State University

Tempe, AZ

Bachelor of Science in Computer Science | **GPA: 4.0/4.0**

Aug. 2022 – May 2026

- **Achievements:** New American University Scholarship, Moeur Award for Academic Excellence, TA of the Year, Dean's List
- **Coursework:** Data Structures and Algorithms, Operating Systems, Software Engineering, Artificial Intelligence, Machine Learning, Natural Language Processing, Database Management, Applied Computational Methods, Cybersecurity

EXPERIENCE

Software Engineering Intern

Tempe, AZ

APMAC Consulting

Aug. 2025 – May 2026

- Built an AI-powered workforce analytics SaaS with **Next.js, FastAPI, and PostgreSQL** to quantify employee productivity in monetary terms, enabling sponsors to evaluate **50+** hiring and workforce scenarios using regression forecasts with **95% confidence intervals**.
- Engineered the **Next.js–FastAPI integration layer**, normalized relational schemas, and built **D3.js sponsor-facing dashboards** for predictions, insights, and reporting, ensuring reliable data flow across frontend, backend, and ML services.
- Developed a HelpDesk ticketing system with **enterprise SSO authentication** and contributed to **AWS deployment**, cloud configuration, and multi-service orchestration, strengthening platform reliability and internal production readiness.

Software Engineer Intern

Remote

Miran Informatics Pvt Ltd

Jun. 2025 – Aug. 2025

- Designed the company's new website using **HTML, Bootstrap, CSS, and JavaScript**, improving the interface and user experience and increasing user traffic by **50%** in the first month after launch.
- Developed **REST API and SQL workflows**, including CRUD services, authentication, role-based access control, reporting/export pipelines, and resilient error handling; reduced production defects by **30%** through regression testing.
- Worked across the software delivery lifecycle for client-facing web applications, translating requirements into production-ready features and supporting debugging, release validation, and version control.

Undergraduate Teaching Assistant

Tempe, AZ

Fulton Schools of Engineering, Arizona State University

Aug. 2023 – May 2026

- Supported **2,000+ students** in CSE 110 and CSE 205 through lab sessions covering object-oriented programming, data structures, and software engineering, helping improve average lab grades by **42%**.
- Led **50+ Teaching Assistants** and developed **Python grading and course-support automation**, reducing manual effort by **70%** and standardizing programming-assignment evaluations.

Undergraduate Researcher

Tempe, AZ

Intelligent Robotics and Interactive Systems (IRIS) Lab, Arizona State University

Jan. 2026 – May 2026

- Researched and developed **MuJoCo reinforcement-learning simulations** for aerial and articulated robotic systems to evaluate policy robustness, environment stability, and sim-to-real readiness under contact-rich dynamics.

PROJECTS

QueueForge | *Next.js, TypeScript, FastAPI, Redis, PostgreSQL, Railway*

- Architected and deployed QueueForge, a cloud-hosted job orchestration platform using Next.js, TypeScript, FastAPI, Redis, PostgreSQL, and Railway, decoupling API handling from background execution through a distributed queue-worker architecture.
- Engineered a Python worker with Redis queue consumption, SQLAlchemy persistence, and detailed execution logging, validating **150+ jobs** with **97% completion** across text-processing and CSV ingestion workloads.
- Built a real-time operations dashboard with Tailwind CSS, shadcn/ui, Zustand, and React Query to expose live job throughput, queue depth, lifecycle state transitions, and failure diagnostics for end-to-end observability and service reliability.

Lofty AI CRM | *LangChain, VAPI, ElevenLabs, DeepSeek*

- **Won the Lofty track at GlobeHack 2026** by building **five AI-powered CRM features**, including an AI voice caller that qualifies leads and books appointments, a DeepSeek control panel, Google Maps route optimization, and SMS alerts.
- Engineered an **SSE-streamed agentic onboarding flow** with LangChain that converts freeform workflow descriptions into a nine-widget dashboard in real time, plus structured-data extraction that automatically books qualified appointments.

SKILLS

Languages: Java, Python, C++, JavaScript, SQL, HTML/CSS, Go

Frameworks & Libraries: React, Next.js, FastAPI, Flask, Node.js, D3.js, SQLAlchemy, React Query, Tailwind CSS

Databases & Infrastructure: PostgreSQL, MySQL, Redis, AWS, Docker, Kubernetes, Railway, Vercel

Tools & Concepts: Git/GitHub, Linux, Jupyter, MuJoCo, REST APIs, distributed systems, message queues, software testing